

13.09.2024. Inductance

13 September 2024 07:58

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\vec{l}_1 \times \hat{r}}{r^2}$$

$$\vec{B} \propto I_1$$

$$\Phi_2 = \int \vec{B}_1 \cdot d\vec{a}_2$$

$$\Phi_2 \propto I_1 \Rightarrow \Phi_2 = M_{21} I_1$$

M_{21} - mutual inductance.

$$M_{21} = M_{12}$$

vector potⁿ, $\vec{\nabla} \cdot \vec{B} = 0$

$$\Rightarrow \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$$

$\hookrightarrow$ vector potential.

Ampere's law, $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$

$$\Rightarrow \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \mu_0 \vec{J}$$

$$\Rightarrow \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J} \quad \text{--- (A)}$$

Vector triple product.

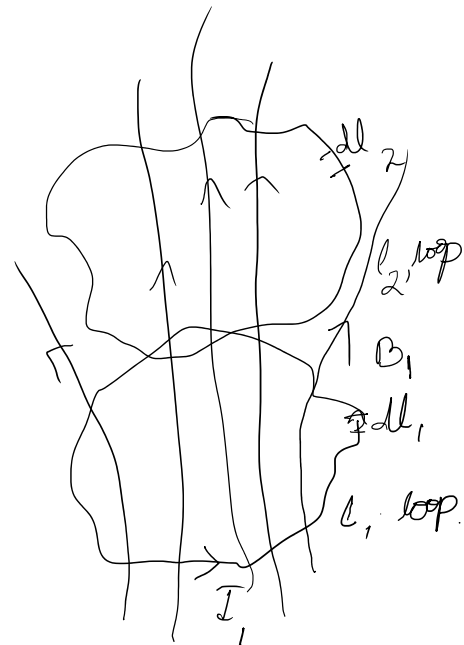
$$\vec{A}' = \vec{A} + \vec{\nabla} f$$

$$\vec{\nabla} \cdot \vec{B}$$

$$\begin{aligned} \vec{B} &= \vec{\nabla} \times \vec{A}' \\ &= \vec{\nabla} \times (\vec{A} + \vec{\nabla} f) \\ &= \vec{\nabla} \times \vec{A} \end{aligned}$$

$$\boxed{\vec{\nabla} \cdot \vec{A} = 0}, \text{ Gauge invariance}$$

$$\hookrightarrow \nabla^2 \vec{A} = -\mu_0 \vec{J} \quad \text{Poisson's eqn.}$$



$$\Rightarrow \boxed{\nabla^2 A = -\mu_0 \vec{J}} \quad \text{Poisson's eqn.}$$

Scalar potential, $\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$.

$$\vec{E} = -\vec{\nabla} V$$

$$\vec{\nabla} \cdot (\vec{E}) = -\vec{\nabla} \cdot (\vec{\nabla} V) = -\nabla^2 V = \rho / \epsilon_0$$

$$\boxed{\nabla^2 V = -\rho / \epsilon_0}$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r')}{r} d\tau'$$

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(r)}{r} d\tau' = \frac{\mu_0}{4\pi} \int \frac{I(r) dl}{r}$$

$$\Phi_2 = \frac{\mu_0}{4\pi} I_1 \oint \oint \frac{dl_1 \cdot dl_2}{r}$$

$$\boxed{M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{dl_1 \cdot dl_2}{r} = M_{12} = M}$$

$$\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -M \frac{dI_1}{dt}$$

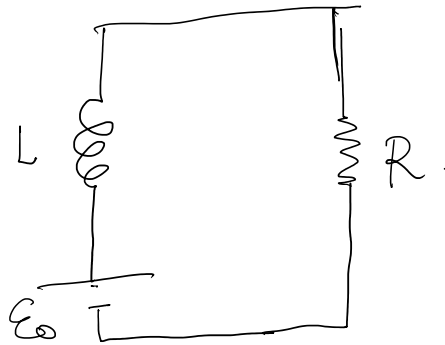
On changing current in loop 1 induces emf in loop 2 along loop 2.

Self inductance, $\Phi = L I$.
 $\downarrow$
 Self inductance.

emf induced due to self inductance,

$$\boxed{\mathcal{E} = -L \frac{dI}{dt}}$$

(H) Henries is the unit for inductance - Volt-second per Ampere.



$$I(t) =$$

problem - 7.20 - 7.25.

Energy in magnetic field :-

$\mathcal{E}$ - work done per unit charge.

work done per unit time,

$$\begin{aligned} \frac{dw}{dt} &= -\mathcal{E} I \\ &= L I \frac{dI}{dt} \end{aligned}$$

$$I = \frac{dq}{dt}$$

Over the time the circuit starts from zero current till I current,

$$\int_0^t \frac{dw}{dt} dt = \int_0^I L I \frac{dI}{dt} dt$$

$$\Rightarrow \boxed{W = \frac{1}{2} L I^2}$$

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{a} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a}$$

$$= \oint \vec{A} \cdot d\vec{l} = LI.$$

$$\Rightarrow W = \frac{1}{2} \oint \vec{A} \cdot d\vec{l}$$

$$= \frac{1}{2} \oint (\vec{A} \cdot \vec{l}) dl$$

$$W = \frac{1}{2} \int_V (\vec{A} \cdot \vec{J}) d\tau$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\Rightarrow \vec{J} = \frac{1}{\mu_0} (\vec{\nabla} \times \vec{B})$$

$$W = \frac{1}{2} \int_V \vec{A} \cdot \left(\frac{1}{\mu_0} \vec{\nabla} \times \vec{B} \right) d\tau$$

Scalar triple product.

$$\Rightarrow W = \frac{1}{2\mu_0} \int B^2 d\tau - \int \vec{\nabla} \cdot (\vec{A} \times \vec{B}) d\tau$$

↓
Gauss's law,

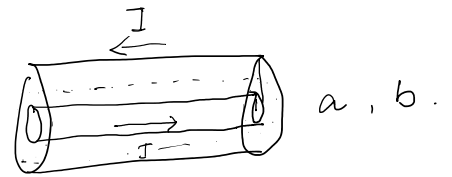
$$= \frac{1}{2\mu_0} \int B^2 d\tau - \int (\vec{A} \times \vec{B}) \cdot d\vec{s}$$

$$\boxed{\begin{aligned} \textcircled{W} &= \frac{1}{2\mu_0} \oint_{\text{larger volume}} B^2 d\tau \\ W &= \frac{\epsilon_0}{2} \oint E^2 d\tau \end{aligned}}$$

Ex 7.13.

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$$B = \frac{\mu_0 I}{2\pi s}$$



Q. What is the mag. field stored.

$$W = \int \frac{B^2}{2\mu_0} d\tau$$

$$W = \int_a^b \frac{B^2}{2\mu_0} 2\pi s l ds$$

What is the self inductance?

$$W = \frac{1}{2} L I^2 = W_{\text{integration}}$$

$$L = \frac{W}{I^2}$$

Maxwell's Equations

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

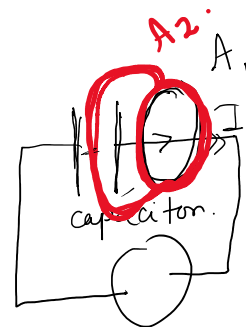
Point 1

Ampere's law,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

$$= \mu_0 I (A_1)$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$



$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$$

$$= 0$$

Point 2 For unsteady current Ampere's law is not valid.

Divergence curl of vector is Zero.

$$0 = \nabla \cdot (\nabla \times \mathbf{E}) = \nabla \cdot \left(\frac{\partial \mathbf{B}}{\partial t} \right) = \frac{\partial}{\partial t} (\nabla \cdot \mathbf{B}) = 0$$

$$\begin{aligned} \nabla \cdot (\nabla \times \mathbf{B}) &= \nabla \cdot (\mu_0 \mathbf{J}) \\ &= \mu_0 (\nabla \cdot \mathbf{J}) \end{aligned}$$

$$\boxed{\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}} \quad \text{Equation of continuity}$$

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$$

$$+ \frac{\partial}{\partial t} (\epsilon_0 \nabla \cdot \mathbf{E})$$

$$\Rightarrow \nabla \cdot \mathbf{J} + \frac{\partial}{\partial t} \epsilon_0 (\nabla \cdot \mathbf{E}) = 0$$

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}}$$

1888 Hertz experimentally Verified.

A changing electric field will introduce a magnetic field.

For loop A2

$$E = \frac{\sigma}{\epsilon_0} = \frac{1}{\epsilon_0} \frac{Q}{A}$$

$$\frac{\partial E}{\partial t} \perp I$$

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} I$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} + \mu_0 \epsilon_0 \int \frac{\partial E}{\partial t} \cdot d\vec{a}$$

$$= \frac{1}{\epsilon_0 A} I \cdot A = \frac{I}{\epsilon_0} \mu_0 \epsilon_0 = \mu_0 I$$

prob. 7.31 - 7.35.

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Displacement current.

Maxwell's equations,

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad - \text{Maxwell's 4th}$$